Physics 4B Spring 2026
Week 1 Discussion

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1. **Topics to be covered by the end of Week 2**

   - Simple Harmonic Motion (Shankar Chapter 17)
     - ◇ Harmonic oscillator
     - ◇ Damped harmonic oscillator (solutions: overdamped, critically damped, under-damped)
     - ◇ Driven damped harmonic oscillator[1] (solutions)
     - ◇ Resonance, Fourier series[2]
   - Waves (Shankar Chapters 18-19)
     - ◇ Wave equation (derivation, solutions)
     - ◇ Wave frequency, period, wavenumber, phase velocity
     - ◇ Wave energy and transmitted power
     - ◇ Doppler effect (derivation)
     - ◇ Interference (double-slit)
     - ◇ Standing waves

   *The highlighted topics have not yet been covered.

2. **Brainwarmer**[3]. For the following pair of numbers, give (i) their polar form, their complex conjugates, their moduli, (ii) their product, and their quotient z_1/z_2:

$$z_1 = 1 + i, \;\; z_2 = \sqrt{3} - i. \tag{1}$$

   Solution:

   (i) $z_1 = \sqrt{2}e^{i\pi/4}$, $z_1^* = 1 - i = \sqrt{2}e^{-i\pi/4}$, $|z_1| = \sqrt{2}$
   $z_2 = 2e^{-i\pi/6}$, $z_2^* = \sqrt{3} + i = 2e^{i\pi/6}$, $|z_2| = 2$

   (ii) $z_1 z_2 = 2\sqrt{2}e^{i\pi/12}$, $z_1/z_2 = \frac{1}{\sqrt{2}}e^{5i\pi/12}$

3. **Simple Harmonic Motion**

[1] We ignore the transient state for the driven oscillator.
[2] Briefly mentioned but not required.
[3] Shankar 8.3

(a) **Damped Harmonic Oscillator.** We have seen from the lecture that the solutions for a damped harmonic oscillator, described by the ordinary differential equation $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$, look like

$$x(t) = Ae^{\alpha_+ t} + Be^{-\alpha_- t} \tag{2}$$

with $\alpha_\pm$ given by

$$\alpha_\pm = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \tag{3}$$

- Prove that for a critically damped harmonic oscillator the solution is the following (Eq. (17.76) of Shankar):

$$x(t) = e^{-\gamma t/2}[A + Bt] \tag{4}$$

Solution: To show that (4) is a solution, we simply plug it into the original ODE

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0 \tag{5}$$

But how do we know that there are two solutions and that the second solution has to look like this?
First of all, we know there has to be two solutions because (i) there are repeated roots for ω when the system is critically damped, and (ii) there are two degrees of freedom in this problem that we usually fix by specifying the initial position $x(0)$ and initial velocity $v(0) = \dot{x}(0)$.
Now, knowing that, we can rewrite the ODE as

$$(D + \gamma/2)^2 x = 0 \tag{6}$$

where $D \equiv \frac{d}{dt}$ and I used the fact that $\omega_0 = \gamma/2$ for a critically damped oscillator. (6) clearly shows that the familiar $Ae^{-\gamma t/2}$ is a solution to the equation $(D+\gamma/2)x = 0$, but it also shows that anything that leads to

$$(D + \gamma/2)x = e^{-\gamma t/2} \tag{7}$$

would be a solution as well.
We will apply a trick here: we already know that $x_1 = e^{-\gamma t/2}$ is a solution, so we can separate it out. As it will turn out, this reduces the problem into something simpler. We write

$$x(t) = x_1(t)u(t) \tag{8}$$

with $x_1(t) = e^{-\gamma t/2}$. If we plug this into (7), the left hand side becomes

$$(D + \gamma/2)x = \dot{x}_1 u + x_1 \dot{u} + \frac{\gamma}{2} x_1 u = x_1 \dot{u} \tag{9}$$

We see that choosing $x_1(t) = e^{-\gamma t/2}$ makes the first and third term cancel. What we are left to do is solve $\dot{u}e^{-\gamma t/2} = e^{-\gamma t/2}$, which leads to $u(t) = t + C$.
Combine this result with (8) and the known solution, we get (4).

- Prove that for a damped oscillator equilibrium is approached fastest for critical damping.

  Solution: Before we start, we must note that we are not considering the under-damped case, which obviously reaches the equilibrium the fastest (but overshoots every time). The envelope of the under-damped oscillation will however decay as $e^{-\gamma t/2}$. Therefore, the question is better phrased as proving that the critical damping approaches the equilibrium fastest *without* overshooting.
  For an overdamped oscillator, the α_+ mode dominates because it decays more slowly than the other, with decay rate (if we write the decay as $e^{-\lambda t}$)

$$\lambda_{\text{over}} = -\alpha_+ = \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \tag{10}$$

  since $\gamma > 2\omega_0$, the square root is positive, so we have $\lambda_{\text{over}} < \frac{\gamma}{2}$.
  For critical damping, the $te^{-\gamma t/2}$ solution dominates, and we can write it as $e^{-\gamma t/2 + \ln t}$. It's not straightforward to convert this directly to a decay rate like we did in the overdamped case, but the $\ln t$ part in the exponent grows very slowly and gets dominated by the $-\gamma t/2$ part, so we can imagine that the effective rate is $\lambda_{\text{crit}} \approx \gamma/2 > \lambda_{\text{over}}$ as $t \to \infty$. Therefore, for sufficiently large t, we should expect $te^{-\gamma t/2} < e^{-\lambda_{\text{over}} t}$.

(b) **Example 1**[4]. A 2 kg mass is connected to a spring with $k = 400$ N/m. The value of the damping coefficient is $\gamma = 1$ s^{-1}. It is driven by a force $F(t) = 10\cos(10t)$ N. What is (i) the amplitude at this ω, (ii) the resonant frequency ω_0, and (iii) the amplitude at resonance?
Solution:

(i) Let's take this opportunity to derive the solution for the driven oscillator:

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos(\omega t) \tag{11}$$

[4]Shankar 8.30

Compared to the damped oscillator, the only change is the driving term on the right hand side. Recall that the solutions for the damped oscillator (derived in lecture) are in general complex exponentials[5]. Using the linearity of (11), we can simplify the problem by solving the following equation instead

$$\ddot{z} + \gamma\dot{z} + \omega_0^2 z = \frac{F_0}{m}e^{i\omega t} \tag{12}$$

Like before, we assume the solution takes the form

$$z(t) = z_0 e^{i\omega t} \tag{13}$$

This is an educated guess (ansatz) because we know that over time, the system has to oscillate at the driving frequency.
Plug (13) into (12) to solve for z_0, and we get the solution for the driven oscillator (in complex form)

$$z(t) = \frac{F_0/m}{-\omega^2 + i\gamma\omega + \omega_0^2}e^{i\omega t} = \frac{F_0}{mZ(\omega)}e^{i\omega t} \tag{14}$$

To find the amplitude, we have to rewrite the solution as $x(t) = \mathrm{Re}z(t) = A\cos(\omega t - \phi)$. Note that in (14) both $Z(\omega)$ and $e^{i\omega t}$ are complex. The easiest way to find the amplitude is to write the complex solution in polar form, in which the amplitude is just the part without the phase factor ($e^{i(\cdots)}$). We start by writing $Z(\omega)$ in polar form

$$Z(\omega) = |Z|e^{i\phi} \tag{15}$$

where $|Z| = \sqrt{(\omega_0^2 - \omega^2)^2 - \omega^2\gamma^2}$, and we do not need to worry about ϕ in this problem. Putting this back into (14), we see the amplitude is $A = \frac{F_0}{m|Z|} = \frac{F_0}{m\sqrt{(\omega_0^2-\omega^2)^2+\omega^2\gamma^2}}$.

(ii) There's some confusion in the question here. As I mentioned during the discussion, the natural frequency $\omega_0 = \sqrt{k/m}$ is not the frequency at resonance (even though the question calls ω_0 the "resonant frequency"). To find the resonant frequency, we have to maximize A, which is equivalent to minimizing

$$(\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2 \tag{16}$$

The solution for $(\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2 = 0$ is the resonant frequency.

[5]The actual solutions are, of course, the real parts of the complex exponentials. We want to use complex exponentials because they are easy to take derivatives with, and the reason we can carry out the whole process with complex exponentials is due to the linearity of the differential equation.

(iii) Once we find the resonant frequency, plug it into the expression for A to get the amplitude.

(c) **Example 2**[6]. For a damped oscillator (not driven by any external force) find the time T^* after which the amplitude of oscillations drops to half its value in terms of γ and m.

Solution: For an oscillator described by the equation $m\ddot{x} + \gamma\dot{x} + kx = 0$[7], the solution decays as $e^{-\gamma t/2m}$. After time T^*, the amplitude drops by half, which means

$$e^{-\gamma T^*/2m} = \frac{1}{2} \tag{17}$$

The solution to this is $2m \ln 2/\gamma$.

4. **Waves**

(a) **Wave Equation**. Derive the wave equation on your own and convince yourself that all terms of order θ^2 are dropped in the process.

Solution: You can find the detailed derivation in Section 18.1 of Shankar, which does a pretty good job. Note that the wave equation only holds for a string if

- The tension and linear density are uniform throughout the string
- The displacement of each string segment from the equilibrium is sufficiently small (compared to the wavelength)
- No damping or driving forces (energy is conserved)
- The string is sufficiently thin compared to the displacement of string elements and the wavelength.

(b) **Standing Waves**.

- Find an expression for all the resonant frequencies of an open-open system of length L.

  Solution: For an open-open system, the fundamental frequency consists of 1 node in the middle and 2 antinodes at the ends, which means $L = \lambda/2 = v/2f$ where v is the wave speed. For each higher mode, we get one extra node and one extra antinode in between, which corresponds to an extra half wavelength, so we have

$$L = \frac{n\lambda}{2} = \frac{nv}{2f_n} \tag{18}$$

[6] Shankar 8.32

[7] Note that in previously discussed examples we have absorbed the m into γ, so the γ here has different dimension and unit.

with $n = 1, 2, 3, \ldots$. Therefore, the frequencies for all the modes are

$$f_n = \frac{nv}{2L}, \; n = 1, 2, 3, \ldots \tag{19}$$

The same result holds for a string with both ends fixed - the only difference is that now the ends have nodes instead of antinodes.

- Find all the resonant frequencies of an open-closed system of length L.

  Solution: For an open-closed system, we have a node at the closed end and an antinode at the open end. The fundamental frequency has exactly one node and one antinode, and therefore $L = \lambda/4$. Each higher mode adds a half wavelength, so

$$L = \frac{n\lambda}{4} = \frac{nv}{4f_n} \tag{20}$$

  with $n = 1, 3, 5 \ldots$. The frequencies are

$$f_n = \frac{nv}{4L}, \; n = 1, 3, 5, \ldots \tag{21}$$

(c) **Doppler Effect**[8]. Let the speed of sound be c_s. A source emitting sound at frequency f_0 is moving at speed u in the $+x$ direction relative to the medium, while an observer is moving at speed v relative to the medium in the $+x$ direction. Show that the equation describing the observed frequency due to Doppler effect is

$$f_{\text{obs}} = f_0 \frac{1 - v/c_s}{1 - u/c_s} \tag{22}$$

Solution: Let's first consider the effect solely due to the motion of the observer by fixing the source. The speed of sound is c_s and the wave fronts are separated by the emitted wavelength $\lambda_0 = c_s/f_0$, but the sound has to catch up to the observer, so it effectively reaches the observer at speed $c_s - v$. The observed frequency is therefore

$$f_{\text{obs, fixed source}} = \frac{c_s - v}{\lambda_0} = f_0 \left(1 - \frac{v}{c_s}\right) \tag{23}$$

Next we consider the effect solely due to the motion of the source by fixing the observer. Moving the source while it emits sound changes the spacings between the wave fronts (emitted at intervals of $t = 1/f_0$), meaning the wavelength gets squeezed or stretched. In this case

$$\lambda = \lambda_0 - u/f_0 \tag{24}$$

[8] Not covered yet

so the observed effect is

$$f_{\text{obs, fixed observer}} = \frac{c_s}{\lambda} = \frac{c_s}{\lambda_0 - u/f_0} = f_0 \frac{1}{1 - \frac{u}{c_s}} \tag{25}$$

Combine the two effects we get the desired result.

(d) **Relativistic Doppler effect**[9]. Show that the expression for the relativistic longitudinal Doppler effect ($v > 0$ means moving away)

$$f_{\text{obs}} = f_0 \sqrt{\frac{1 - v/c}{1 + v/c}} \tag{26}$$

reduces to the non-relativistic version when $v \ll c$.

Solution: With $v/c \ll 1$, we can expand the original expression to get

$$f_{\text{obs}} \approx f_0 \frac{1 - \frac{v}{2c}}{1 + \frac{v}{2c}} \approx f_0(1 - v/2c)(1 - v/2c) \approx f_0(1 - v/c + \mathcal{O}(v^2/c^2)) \tag{27}$$

This is the non-relativistic result for a moving observer.

(e) **Example 3**[10]. (i) Write an equation in MKS units for a wave moving along $+x$ with amplitude 0.4 m, speed 6 m/s , and frequency 17 Hz. (ii) If these are waves on a string with mass per unit length $\mu = 0.02$ kg/m, what is u, the energy per unit length? (iii) What is the average power being fed into the vibrating string?

Solution:

(i) $y(x,t) = A\cos\left(\frac{2\pi f}{v}x - 2\pi f t\right)$

(ii) The energy per unit length is $u = \frac{1}{2}\mu\omega^2 A^2$ - find this result by finding the kinetic energy using $\dot{y}$ and the potential energy using $\partial y/\partial x$ and then average the square sine/cosine over a full period.

(iii) The average power is $P = uv$, by dimensional analysis.

(f) **Example 4**[11]. Find the maximum acceleration of points on a string supporting a standing wave with $v = 16$ m/s, a spacing of 2 m between nodes, and an amplitude of displacement of 2 cm.

Solution: We are given all the quantities we needed to describe the solution of the standing wave

$$y(x,t) = A\sin(kx)\cos(\omega t) \tag{28}$$

[9] This is only intended as an exercise for series expansion - You do not need to know any relativistic effect in this course.

[10] Shankar 9.2

[11] Shankar 9.6

To find the acceleration, take its second time derivative

$$\ddot{y} = -\omega^2 A \sin(kx) \cos(\omega t) \tag{29}$$

The maximum acceleration is

$$a_{\max} = \omega^2 A = (vk)^2 A = v^2 A \frac{4\pi^2}{\lambda^2} \tag{30}$$

Note that the spacing between nodes is half a wavelength.